

BALAVIGNESH SHANMUGAM

AI/ML Engineer • Data Scientist • Blockchain Developer

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LeetCode: [Balavignesh Shanmugam](#) • GFG: [Balavignesh Shanmugam](#) (SSEC) • SkillRack: [#441044](#) • StrataScratch: [Active](#)

EDUCATION

Sri Sairam Engineering College, Chennai

2023 – 2027

B.Tech. Artificial Intelligence & Data Science

CGPA: 8.4 / 10.0

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Blockchain Technology, Data Structures & Algorithms, Database Systems.

TECHNICAL SKILLS

AI / ML / DL	Python, PyTorch, TensorFlow, TensorFlow Lite, Scikit-learn, OpenCV, YOLO v8, Pandas, NumPy, Matplotlib
LLMs & GenAI	Gemini API, Grok API, Perplexity API, LangChain, Multi-Agent Orchestration, RAG Pipelines, Hugging Face, FAISS
Blockchain	Solidity, Ethereum, Web3.py, GOAT Network, AI Wallet Agents, Smart Contracts, dApps, Hot/Cold Wallet Logic
Data Science	EDA, Feature Engineering, Statistical Analysis, Data Visualization, SQL, Sports Analytics, StrataScratch
Backend & APIs	Flask, FastAPI, REST APIs, PostgreSQL, TypeScript, Expo (React Native), Twilio API, SQLAlchemy
DevOps & Cloud	Git, GitHub, AWS (EC2, S3, IAM), Docker (fundamentals), GitHub Actions, Linux — Ubuntu
Tools	Jupyter, VS Code, Postman, Roboflow, Streamlit, Telegram Bot API, OpenClaw SDK, Weights & Biases

EXPERIENCE

Infosys Springboard

Dec 2025 – Feb 2026 | 3 months

AI Engineering Intern Remote

- Designed and deployed an **end-to-end AI wildlife protection backend**
- Using YOLO object detection and Gemini Vision API for poaching activity identification from ranger-uploaded media; multi-stage verification pipeline significantly reduced false positives.
- Built **secure REST APIs with Flask and PostgreSQL**
- Featuring automated real-time alert dispatch, integrating CV inference, backend orchestration, and on-ground conservation response into a single production-grade system.
- Collaborated cross-functionally on system architecture, API documentation, and scalability planning for a conservation-critical AI deployment.

Big Bucks Innovation

Jun 2024 – Jul 2024 | 2 months

AI/ML Intern Remote

- Executed end-to-end ML pipelines covering data preprocessing, feature engineering, model training, and evaluation on real-world datasets; applied industry best practices for model lifecycle management.
- Trained on supervised/unsupervised algorithms, CNNs, and deep learning architectures through structured project-based learning alongside a cross-functional team.

Sri Sairam Techno Incubator Foundation (STIC)

Mar 2024 – Apr 2024

Research and Development Intern Chennai

- Collected and preprocessed real-time research datasets, performed EDA and derived insights to support decision-making for inclusion programs at the incubator drive student engagement with rural communities.

PROJECTS

VORTUX AI — Football Match Analysis Platform

Python · YOLO · Deep Learning · OpenCV · Flask · Web App

- ▶ Built a full-stack AI-powered VAR system using custom YOLO models for player detection, ball tracking, foul recognition, and offside analysis.
- ▶ Developed an end-to-end ML pipeline: data collection, annotation (Roboflow), model training, evaluation, and deployment.
- ▶ Working on team analytics dashboard with heatmaps, possession stats, and match timelines for tactical insights.

LifeGuard AI — Elderly Care Mobile Application

TypeScript · Expo · TensorFlow Lite · Gemini · Grok · Twilio

- ▶ Developed a cross-platform mobile app for elderly care with on-device ML and cloud-based AI integration.
- ▶ Implemented real-time fall detection using TensorFlow Lite with automated emergency alerts and location sharing via Twilio.
- ▶ Built AI health features including a multi-LLM chatbot and image-based skin condition detection.

GOAT Wallet Agent — Autonomous Blockchain AI Agent |

Python · GOAT Network · OpenClaw · Telegram Bot API · Web3

- ▶ Engineered an autonomous blockchain AI agent for fund management using real-time market and on-chain signals.
- ▶ Designed a decision engine for portfolio rebalancing based on price trends, liquidity, and whale activity.
- ▶ Integrated a Telegram bot interface for natural language control and real-time portfolio monitoring.

PurePath AI — Multi-Agent Water Risk Orchestration System |

Python · LangChain · Multi-Agent · NLP · Government API Integration

- ▶ Built a multi-agent system using LangChain to predict water scarcity risks in rural communities.
- ▶ Designed specialized agents for risk prediction, stakeholder mapping, and government scheme matching.
- ▶ Enabled data-driven alerts connecting at-risk communities with NGOs and public water initiatives.

CERTIFICATIONS

- ▶ Introduction to Machine Learning NPTEL / IIT
- ▶ Programming, Data Structures & Algorithms using Python NPTEL / IIT
- ▶ Data Analytics with Python NPTEL / IIT
- ▶ Python for Data Science NPTEL / IBM
- ▶ Certification in Risk Management Assurance (CRMA) Professional Body
- ▶ Data Science using Python International Workshop Brainovision Solutions × Sri Sairam EC | Feb 2024

COMPETITIVE PROGRAMMING & PROFILES

LeetCode [Balavignesh Shanmugam](#) - problem solver - Data Structures & Algorithms

GeeksForGeeks GFG: [Balavignesh Shanmugam](#) (SSEC) - Consistent contributor

SkillRack [skillrack.com/profile/441044](#) - College-ranked profile

StrataScratch [Balamyself](#) - SQL + Python data science centric problems

LEADERSHIP & ACTIVITIES

- **Operations Lead (Mastermind), MAGIC Team - Unnat Bharat Abhiyan** 2025 – 2026
Led rural development initiatives impacting 10 villages; managed events, student-community programs, and cross-departmental coordination.
- **Village Communicator, Core Team Somangalam - UBA** 2024
Coordinated direct community engagement and rural upliftment programs as core team member.
- **Student Volunteer - UBA Sairam Chennai Chapter** 2023